

PROJECT IA

TENNIS MATCH PREDICTION

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The application of **Artificial Intelligence** to sports analytics has gained significant attention in recent years. **Predicting sports** outcomes is challenging due to uncertainty, variations in player performance, and external factors affecting matches. **Tennis**, with its large amount of historical data, provides a suitable domain for predictive modeling. The objective of this project is to develop a machine learning model capable of **predicting the winner of a tennis match** using historical match data as a binary classification problem.

The **dataset** used in this project is sourced from publicly available records provided by Tennis-Data, which include detailed information about matches across multiple years. These datasets contain various attributes such as **player rankings, match statistics, tournament details**, and **betting odds**, offering a comprehensive basis for analysis.

To complete this project, several steps were necessary. First, the **raw data is preprocessed** and enriched through **feature engineering** to extract important indicators of player performance, like ranking differences and recent form. Then, multiple **machine learning models are trained** and evaluated. In particular, the project compares a baseline model implemented using “scikit-learn” with a more advanced gradient boosting approach, allowing for a deeper **analysis** of predictive performance and model behavior.

Through this approach, the project aims not only to achieve accurate predictions but also to better understand the key factors influencing match outcomes.

Dataset Description :

The dataset used in this project is obtained from Tennis-Data, which provides detailed records of professional tennis matches in the form of yearly Excel files.

Overall, the dataset includes a large number of matches (typically several thousand per year), with each match described by numerous features capturing information about the players, the match context, and pre-match expectations. These features allow for a comprehensive analysis of the factors influencing match outcomes.

The variables in the dataset can be grouped into several categories: **Player information** (names of the winner & loser, handedness), **ranking data** (ATP rankings and ranking points of both players at the time of the match), **match characteristics** (tournament, surface type, round, and date of the match), **match statistics** (performance indicators like aces, double faults, first serve percentage, and other in-game statistics) and **betting odds** (pre-match odds, market expectations about the likely winner).

This diversity of features provides a rich basis for building predictive models, as it combines objective performance metrics with contextual and probabilistic information.

Data Cleaning :

The raw dataset required several preprocessing steps to ensure its quality and usability for the models. Missing values were handled by removing rows containing incomplete or inconsistent information, particularly when features like rankings or betting odds were missing. Additionally, irrelevant or redundant columns—such as identifiers, textual descriptions, or features not available before the match—were removed to prevent data leakage and reduce noise in the model.

Feature Engineering :

To **improve the predictive performance** of the models, **features were engineered** from the raw dataset to better represent the relative strength of the two players in each match.

First, the dataset was **restructured** so that each match is shown as a symmetric "**Player 1 vs Player 2**" format, with a random assignment of players to avoid bias. A binary label indicates if Player 1 wins.

The main engineered features are based on differences between the two players. In particular, the **ranking difference** and **points difference** were computed to capture the skill gap between players. These features are strong indicators of performance, as higher-ranked players are generally more likely to win.

Then, **betting odds** were transformed into implied probabilities (1/odds) for each player, and their difference was used as a feature. This is interesting, as the betting odds comes from a large amount of expert and their statistical knowledge about match outcomes.

And finally, **categorical variables** describing the match context (surface, tournament, court type, and match format) were changed into numerical values to be useful. These features allow the model to account for contextual factors that can influence performance.

Overall, the feature engineering process focuses on transforming raw match data into **meaningful numerical differences** and contextual indicators, so the models can better distinguish between the two players.

Models :

To predict the outcome of tennis matches, two different machine learning models were implemented and compared. The objective was to evaluate both a classical model and a more advanced boosting method.

– Decision Tree (Scikit-learn) :

The first model used is a Decision Tree Classifier implemented with "scikit-learn". It was chosen as a baseline due to its simplicity and interpretability. Decision trees work by **recursively splitting the data** based on feature values in order to **maximize the separation between classes**.

Key parameters such as the **maximum depth of the tree** and **minimum samples per leaf** were tuned to avoid overfitting and improve generalization. The model is particularly useful for understanding how different features affect the prediction, as it provides a clear structure of decision rules.

– LightGBM (beyond Scikit-learn) :

The second model is based on LightGBM, a powerful gradient boosting algorithm designed for efficiency and high performance. Unlike decision trees, LightGBM builds an **ensemble of trees sequentially**, where each new tree focuses on correcting the errors made by the previous ones.

This model was selected because it is well-suited for structured data and often achieves better predictive performance. Several hyperparameters, like the number of estimators, learning rate, and leaves, were tuned to optimize the model.

Compared to the Decision Tree, LightGBM is able to capture more complex relationships between features, which can lead to improved accuracy in predicting match outcomes.

Metrics :

To evaluate the performances, **accuracy** was used as the main metric due to the balanced binary nature of the task.. It measures the **proportion of correctly predicted match outcomes over the total number of predictions**. This metric is good for this task since it is a balanced binary classification problem (predicting the winner between 2 players).

RESULTS COMPARAISON

The performance of the two models is summarized below:

- **Decision Tree** (scikit-learn, GridSearchCV tuned): **70.3%** accuracy
- **LightGBM** (GridSearchCV tuned): **70.7%** accuracy

The results show that both models achieve very similar performance, with LightGBM slightly outperforming the Decision Tree.

The Decision Tree already achieves **relatively strong performance**, suggesting that the engineered features (like ranking and probability differences) are highly informative and allow a simple model to make good predictions. Surprisingly, the LightGBM model only provides a **small improvement** over the Decision Tree. This shows that **the problem may not require complex models**, as most of the predictive power is captured by a few key features. Those derived from rankings and betting odds likely dominate the prediction, leaving little room for more advanced models to improve performance.
Another possible explanation is that tennis **match outcomes contain a degree of inherent uncertainty**, due to the player condition, injuries, or psychological aspects, which are not captured in the dataset. This **limits the maximum achievable accuracy** regardless of the model used.

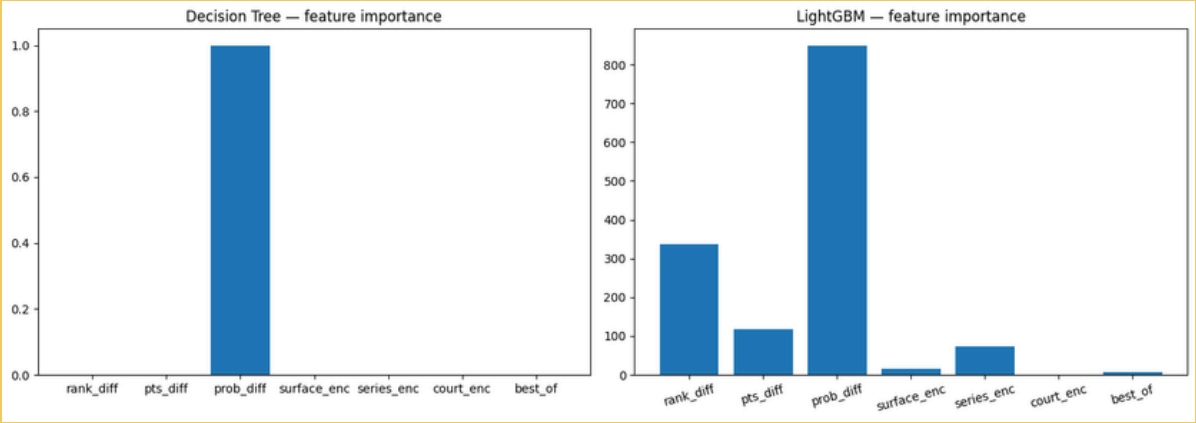
To gain a more precise understanding of model performance, additional evaluation metrics were considered, like **precision, recall, F1-score, and AUC-ROC** - results may vary slightly due to the random restructuring step. Both the Decision Tree and LightGBM models show very similar results across all metrics. In particular, **precision and recall are well balanced** for both classes (around 0.70–0.72), indicating that the models do not favor one class over the other. The F1-score, which combines precision and recall, is also consistent at approximately 0.71 for both models.
The **AUC-ROC score** provides additional insight into the models' ability to discriminate between the two classes. The **Decision Tree achieves an AUC of 0.765**, while **LightGBM slightly improves this score to 0.767**. Although the improvement is small, it confirms that **LightGBM has a slightly better capacity to rank predictions correctly**.

Overall, these results reinforce the conclusion that **both models perform similarly**, with LightGBM offering only marginal improvements. The consistency across multiple evaluation metrics suggests that the **models are stable** and that their performance is not dependent on a single measure.

Feature Importance :
For the **Decision Tree**, the results show a strong **dominance of the probability difference** feature, which accounts for **almost all of the model's decision-making process**. The other features, like ranking difference and points difference, have little impact. This shows that the Decision Tree relies almost entirely on betting odds to make predictions, using it as a primary decision rule.
For the LightGBM model, the **feature importance is more distributed**, although **probability difference remains the most important feature** by a large margin. But, point difference and ranking difference contribute significantly to the predictions, while contextual features (surface, series, and court type) have a smaller influence. This shows that **LightGBM is able to leverage a broader set of features** and **capture more complex relationships** compared to the Decision Tree.

Overall, **both models shows the dominant role of betting odds** in predicting match outcomes. This is expected, odds reflect the knowledge of expert and statistical information. But the results also show that additional features provide complementary information that improves prediction performance, especially in the more advanced LightGBM model. So, while a single strong feature can already yield good predictions, combining multiple relevant features allows for a more robust and nuanced model.

MOST INFLUENTIAL FEATURES FOR MATCH PREDICTION



Baseline comparaison :
Two baselines were established to contextualise model performance. The majority class baseline confirm the near-balanced label distribution from the coin flip restructuring.
More critically, the bookmaker baseline — always predicting the player with higher implied probability — confirm that betting market odds alone are an extremely strong predictor. Both models match this baseline precisely, with no meaningful improvement over simply following the bookmaker. This validate that prob_diff already encodes virtually all predictable information about match outcomes.
Genuine improvement would require features that capture information not yet priced into the odds — such as player fatigue, injury status, or surface-specific recent form computed from historical data.

Baselines :
Majority class accuracy: 0.532
Bookmaker accuracy: 0.707
Models :
DT accuracy: 0.707
LightGBM accuracy: 0.708

Improvement over majority class: +0.175
Improvement over bookmaker: +0.000

Error Analysis :
To better understand the model, an analysis of incorrect predictions was conducted using match context features like surface type and tournament category. The exact number of errors changes slightly between executions because some preprocessing and train/test splitting steps involve randomness.

Analysis by Surface : The error rates are **relatively close between the two surfaces** (approximately 30% each), suggesting that **surface type alone does not significantly influence the model's performance**.

Analysis by Tournament Series : The analysis of tournament categories shows **more noticeable differences**. The largest number of errors is in ATP250 tournaments (94 errors) and Masters 1000 tournaments (79 errors). These categories also **present the highest error rates (**32.4% for ATP250 and 33.1% for Masters 1000 events).

In contrast, Grand Slam tournaments show the lowest error rate at 22.5%, suggesting that these matches are generally easier to predict.
One possible explanation is that top-ranked players tend to perform more consistently during Grand Slam events, leading to clearer differences between competitors.

Interpretation : Overall, the error analysis confirm the intuition that **the model struggles more in tournaments where matches are highly competitive** and player levels are closer.
On the other hand, matches involving stronger favorites, such as those often found in Grand Slam tournaments, are predicted more accurately.
These results highlight an important limitation of sports prediction models: even with relevant features and advanced algorithms, match outcomes remain difficult to predict when the level difference between players is small.

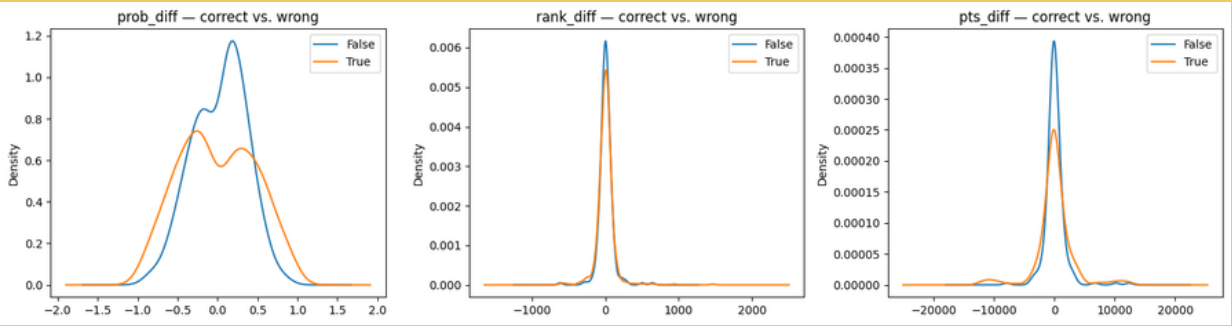
And in the graph below, we see that **wrong predictions concentrate around prob_diff ≈ 0**, evenly matched matchs where the bookmaker sees no clear favourite. rank_diff and pts_diff show identical distributions for correct and wrong predictions, confirming these features do not explain model failures.

EXAMPLE OF PREDICTION ERRORS:

Wrong predictions by surface:
Hard : 187
Clay : 63
Wrong predictions by series:
ATP250 : 94
Masters 1000 : 79
ATP500 : 50
Grand Slam : 27

Error rate by surface:
Hard : 0.301613
Clay : 0.297170
Error rate by series:
Masters 1000 : 0.330544
ATP250 : 0.324138
ATP500 : 0.273224
Grand Slam : 0.225000

FEATURE DISTRIBUTION: CORRECT VS. WRONG PREDICTIONS



Conclusion :

This project demonstrates that tennis match outcomes can be predicted with **reasonable accuracy** (~71%) using a **small set of well-engineered features**. Betting odds and player rankings provide most of the predictive power, while the comparison between Decision Tree and LightGBM shows that model complexity has only a limited impact on performance.
The analysis also highlights the challenges of sports prediction, as **the models struggle with closely contested matches** where outcomes are inherently uncertain. Since betting odds already encode most predictable signals and factors like injuries or fatigue are missing, **some matches remain fundamentally hard to model**.
Overall, the project shows both the potential and the limits of machine learning for predicting tennis results.